# CPSC 471 - Progress Report 1 - EERD

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# ASSUMPTIONS

**People and Specializations**

- We are assuming that every PERSON entity in the system has a unique Person ID.
- A person is specialized into three overlapping specializations: VOLUNTEER, VICTIM, and RESPONDER. Any given person can be none, one, or more of these specializations.
    - e.g., a volunteer can end up being a victim
- A person can have zero or more MEDICAL INFO records. Making MEDICAL INFO a weak entity, with its existence being solely reliant on a person existing.
- MEDICAL INFO is identified by its Info Number, which is a partial key, and its Person ID.
- We're assuming that every VICTIM can be identified and assigned a Person ID
- The "Has Vehicle" and "Availability Status" attributes for VOLUNTEER are set to either True or False
- "Address" will be set to their home address

**Agencies and Resource Owners**

- AGENCY and RESOURCE OWNER are linked through a union category. This means we are assuming a resource owner can be an agency or a volunteer.
- We are assuming that agencies employ responders. One AGENCY can employ many RESPONDERS.
- AGENCIES operate RESPONSE TEAMS. One AGENCY can operate many teams.
- Agencies are identified by an Agency ID
- Since AGENCY and VOLUNTEER have unrelated primary keys that could hold the same value, we have "Owner_ID" as the primary key for Resource Owner
- Every RESOURCE OWNER must correspond to exactly one AGENCY or VOLUNTEER

**Response Teams**

- Every RESPONSE TEAM has a unique Team ID
- RESPONDERS are members of response teams. Many RESPONDERS can be members of many teams.
- Each RESPONSE TEAM has one lead responder. One responder can lead one response team.
- Team location and size are tracked as current values and can be subject to change as an emergency evolves.

**Resources**

- A RESOURCE is a disjoint specialization, where every RESOURCE can be either a VEHICLE or EQUIPMENT but not both
- Each RESOURCE has a unique Resource ID
- Each RESOURCE is owned by one RESOURCE OWNER. A RESOURCE OWNER can own many resources.
- RESOURCES are used by DISPATCH ASSIGNMENTS. Many RESOURCES can be used by many DISPATCH ASSIGNMENTS.

**System Users**

- SYSTEM USERS are a disjoint specialization into: ADMIN, DISPATCHER, and CLIENT. Each user can only be one of the three.
- Each user has a unique User ID.
- One SYSTEM USER belongs to exactly one PERSON, and every SYSTEM USER has to be a person.
- Many ADMINS manage many AGENCIES.

**Emergencies**

- Each EMERGENCY has a unique Emergency ID
- EMERGENCIES are handled through DISPATCH ASSIGNMENTS. One EMERGENCY can have multiple assignments, while each assignment is related to one EMERGENCY.
- Many VOLUNTEERS can be assigned to many EMERGENCIES
- Each HELP REQUEST is about exactly one EMERGENCY. One EMERGENCY can have many HELP REQUESTS.
- An EMERGENCY has many EVACUATION ZONES and many STATUS UPDATES.

**Dispatch Assignments**

- Each assignment has a unique Assignment ID
- Each assignment is created by a DISPATCHER. A DISPATCHER can create many assignments, and an assignment can be created by exactly one DISPATCHER.
- Each assignment is assigned to a RESPONSE TEAM. A RESPONSE TEAM can receive many assignments, and an assignment can be assigned to one RESPONSE TEAM.
- Dispatchers work for AGENCIES. Many dispatchers can work for one AGENCY.
- ETA is an estimate based on the distance of the RESPONSE TEAM from the specified EMERGENCY

**Help Requests**

- CLIENTS submit help requests. One CLIENT can submit many requests.
- Each request has a unique Request ID.

**Evacuation Zone**

- An EVACUATION ZONE is a weak entity, meaning that it is entirely dependent on the type of EMERGENCY that is occurring in that area. No EMERGENCY means no EVACUATION ZONE.
- Each EVACUATION ZONE has a partial ZONE ID.
- VICTIMS can be located in EVACUATION ZONES. One EVACUATION ZONE can have many VICTIMS located in it.

**Status Updates**

- A single EMERGENCY can have one or multiple STATUS UPDATES. The existence of a STATUS UPDATE depends solely on the EMERGENCY, meaning that if there is no EMERGENCY, there is no status for it, making STATUS UPDATE a weak entity.
- Each STATUS UPDATE has a partial key UPDATE SEQ.

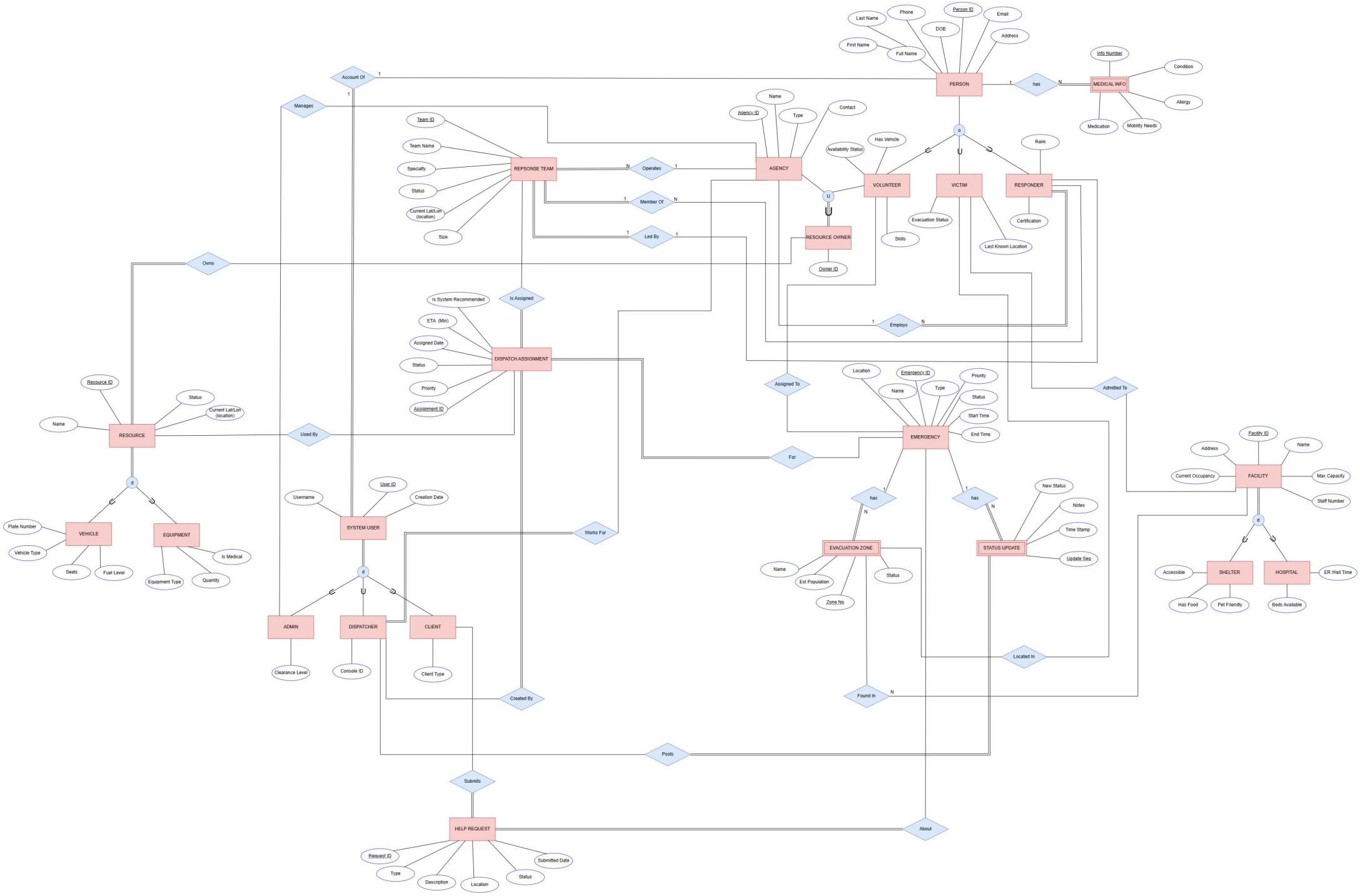
**Facilities**

- EVACUATION ZONES may or may not have FACILITIES found in the same area. When we talk about FACILITIES, we mean either SHELTERS or HOSPITALS. Therefore, the superclass FACILITY has two subclasses, HOSPITAL and SHELTER, and these subclasses are disjoint, meaning a FACILITY is either a HOSPITAL or a SHELTER, but not both.
- Capacity is just a number; it doesn't account for the number of rooms, wards, or different beds.
- A FACILITY can be found in an EVACUATION ZONE. A zone may have many FACILITIES in it.
- VICTIMS can be admitted to FACILITIES. Many VICTIMS can be admitted to one FACILITY, and a VICTIM can be admitted to different FACILITIES over time.
- A FACILITY cannot exceed its maxim capacity.

**Others**

- We assume the data updated in the database is accurate and current.

(Note: Attached is a PNG of our EER; if, for some reason, it is too small or the quality is not good enough since it is in a PDF for the grader to grade, please reach out to luis.hernandez1@ucalgary.ca for the raw PNG for better quality.)


Last Name
Phone
Person ID
Email
DOB
Address
First Name
Full Name
PERSON
Info Number
Condition
Allergy
MEDICAL INFO
Medication
Mobility Needs
Account Of
1
1
has
N
Manages
Name
Agency ID
Type
Contact
Team ID
Team Name
Specialty
Status
Current Lat/Lon (location)
Size
REPSONSE TEAM
Operates
AGENCY
Availability Status
Has Vehicle
Rank
o
VOLUNTEER
VICTIM
RESPONDER
Member Of
U
Led By
RESOURCE OWNER
Owner ID
Evacuation Status
Certification
Skills
Last Known Location
Owns
Is System Recommended
Is Assigned
ETA (Min)
Assigned Date
Status
Priority
Assignment ID
DISPATCH ASSIGNMENT
Employs
Assigned To
Admitted To
Location
Emergency ID
Name
Type
Priority
Status
Start Time
End Time
EMERGENCY
Resource ID
Status
Current Lat/Lon (location)
Name
RESOURCE
Used By
For
d
Facility ID
Address
Name
Current Occupancy
FACILITY
Max Capacity
Staff Number
Plate Number
VEHICLE
EQUIPMENT
Vehicle Type
Seats
Fuel Level
Is Medical
Equipment Type
Quantity
Username
User ID
Creation Date
SYSTEM USER
Works For
has
has
New Status
Notes
Time Stamp
Update Seq
EVACUATION ZONE
STATUS UPDATE
Name
Est Population
Zone No
Status
d
Accessible
SHELTER
HOSPITAL
ER Wait Time
Has Food
Pet Friendly
Beds Available
ADMIN
DISPATCHER
CLIENT
Clearance Level
Console ID
Client Type
Created By
Located In
Found In
N
Posts
Submits
HELP REQUEST
About
Request ID
Type
Description
Location
Status
Submitted Date